Task 4

R code:

```
# Get current working directory

getwd()

#read dataset

data <- read.csv("/Users/apple/Downloads/INRTU courses/2nd semester/Data
Analysis/Homework 3/data_for_analysis.csv", header=T)

install.packages("coin")

install.packages("pROC")

library(coin)

library(pROC)

data$outcome <- as.factor(data$outcome)

summary(data)

> sum(is.na(data$lipids5))

[1] 276

> mean(is.na(data$lipids5)) * 100

[1] 24.04181
```

1. Performing correlation analysis between other variables

R code:

```
# Define variable groups

lipid_vars <- c("lipids1", "lipids2", "lipids3", "lipids4", "lipids5")

hormone_vars <- c("hormone1", "hormone2", "hormone3", "hormone4",

                  "hormone5", "hormone6", "hormone7", "hormone8",

                  "hormone10_generated")

antioxidant_vars <- c("antioxidant1", "antioxidant2", "antioxidant3",

                      "antioxidant4", "antioxidant5")

lipid_pero_vars <- c("lipid_pero1", "lipid_pero2", "lipid_pero3",

                     "lipid_pero4", "lipid_pero5")
```

```
all_vars <- c(lipid_vars, hormone_vars, antioxidant_vars, lipid_pero_vars)


# Testing for normality of all variables
for(var in all_vars) {
  cat("\nNormality test for", var, "\n")
  print(shapiro.test(data[[var]]))


  hist(data[[var]], col = "lightblue",
       main = paste("Histogram of", var))
  qqnorm(data[[var]], main = paste("Q-Q Plot of", var))
  qqline(data[[var]], col = "red", lwd = 2)
}
# Spearman's correlation test between all pairs
cat("\nSpearman Correlation between all variables:\n")
for(i in 1:(length(all_vars)-1)) {
  for(j in (i+1):length(all_vars)) {
    result <- cor.test(data[[all_vars[i]]], data[[all_vars[j]]], method =
"spearman")
    cat(sprintf("%s vs %s: rho = %.3f, p = %.4f\n",
                all_vars[i], all_vars[j], result$estimate, result$p.value))
  }
}
```

2. Obtaining a table with correlation coefficients with significance assessment (permutation method)

R code:

```
# Data frame for results
results <- data.frame(
  var1 = character(),
```

```
  var2 = character(),
  spearman_corr = numeric(),
  s_p_value = numeric(),
  stringsAsFactors = FALSE
)

# Function for permutation-based Spearman correlation (replaces wPerm)
perm_spearman <- function(x, y, method = "spearman", R = 10000) {
  valid <- complete.cases(x, y)
  x_clean <- x[valid]
  y_clean <- y[valid]

  obs_cor <- cor(x_clean, y_clean, method = method)

  perm_cors <- numeric(R)
  for(i in 1:R) {
    perm_cors[i] <- cor(x_clean, sample(y_clean), method = method)
  }

  p_value <- mean(abs(perm_cors) >= abs(obs_cor))

  return(list(Observed = obs_cor, p.value = p_value))
}

# Main loop for all pairs
for(i in 1:(length(all_vars)-1)) {
  for(j in (i+1):length(all_vars)) {
    perm_result <- perm_spearman(
      x = data[[all_vars[i]]],
      y = data[[all_vars[j]]],
```

```
      method = "spearman",
      R = 10000
    )

    results <- rbind(results, data.frame(
      var1 = all_vars[i],
      var2 = all_vars[j],
      spearman_corr = perm_result$Observed,
      s_p_value = perm_result$p.value
    ))
  }
}

# Output result
print("QUESTION 2: Correlation Coefficients with Permutation Significance (ALL VARIABLES)")
print(results)

# Filter significant results
sig_results <- results[results$s_p_value < 0.05, ]
print("Significant Correlations (p < 0.05):")
print(sig_results)
```

3. Performing regression analysis between other variables

R code:

```
# Remove NAs first
df <- data[, c(x_var, y_var)]
df <- na.omit(df)
df <- df[order(df[[x_var]]), ]

# Linear regression
```

```
model_linear <- lm(as.formula(paste(y_var, "~", x_var)), data = df)


# Second degree polynomial

model_2 <- lm(as.formula(paste(y_var, "~ poly(", x_var, ", 2)")), data = df)


# Third degree polynomial

model_3 <- lm(as.formula(paste(y_var, "~ poly(", x_var, ", 3)")), data = df)


# Exponential dependence (y = a * e^(b*x))

# Correct: log(y) ~ x

model_exp <- lm(as.formula(paste("log(", y_var, ") ~", x_var)), data = df)


# Log dependence (y = a + b*log(x))

# Correct: y ~ log(x)

model_log <- lm(as.formula(paste(y_var, "~ log(", x_var, ")")), data = df)
```

4. Selecting the best model (BIC)

R code:

```
rezult <- data.frame(

  model = c("model_linear", "model_2", "model_3", "model_exp", "model_log"),

  BIC_value = c(BIC(model_linear), BIC(model_2), BIC(model_3),

                BIC(model_exp), BIC(model_log))

)


rezult <- rezult[order(rezult$BIC_value), ]

print(rezult)
```

Best Model Selection (BIC):

Based on BIC comparison, model_linear has the lowest BIC value (-6964.082), indicating it is the best fitting model among all tested models. model_2 and model_3 have similar BIC values, suggesting polynomial models do not substantially improve fit over the linear model. model_exp and model_log have much higher BIC values, indicating poor fit. The linear model is selected as the best model for describing the relationship between the variables.

5. Fiting logistic regression models using hormone variables to predict the binary outcome, compare model performance via AIC/BIC, compute odds ratios, and summarize the results

R code:

```
# Fit logistic regression models using hormone variables to predict binary
outcome,

# compare model performance via AIC/BIC, compute odds ratios, and summarize
results.



# Check for NAs

sum(is.na(data$outcome))

data <- data[!is.na(data$outcome), ]



# Create complete dataset (remove rows with any missing values in all
variables)

data_complete <- data[complete.cases(data[, all_vars]), ]



# Model 1: Hormones only

model_logit_hormones <- glm(outcome ~ hormone1 + hormone2 + hormone3 +
hormone4 +

                              hormone5 + hormone6 + hormone7 + hormone8 +

                              hormone10_generated,

                            data = data_complete, family = binomial)

summary(model_logit_hormones)



# Model 2: Lipids only

model_logit_lipids <- glm(outcome ~ lipids1 + lipids2 + lipids3 + lipids4 +
lipids5,

                          data = data_complete, family = binomial)

summary(model_logit_lipids)
```

```
# Model 3: Antioxidants only

model_logit_antioxidants <- glm(outcome ~ antioxidant1 + antioxidant2 +
antioxidant3 +

                                  antioxidant4 + antioxidant5,

                                data = data_complete, family = binomial)

summary(model_logit_antioxidants)



# Model 4: Lipid peroxidation only

model_logit_lipid_pero <- glm(outcome ~ lipid_pero1 + lipid_pero2 +
lipid_pero3 +

                                lipid_pero4 + lipid_pero5,

                              data = data_complete, family = binomial)

summary(model_logit_lipid_pero)



# Model 5: All variables combined

model_logit_all <- glm(outcome ~ .,

                       data = data_complete[, c("outcome", all_vars)],

                       family = binomial)

summary(model_logit_all)



# Compare models via AIC/BIC

model_comparison <- data.frame(

  Model = c("Hormones", "Lipids", "Antioxidants", "Lipid_Peroxidation",
"All_Variables"),

  AIC = c(AIC(model_logit_hormones), AIC(model_logit_lipids),

          AIC(model_logit_antioxidants), AIC(model_logit_lipid_pero),

          AIC(model_logit_all)),

  BIC = c(BIC(model_logit_hormones), BIC(model_logit_lipids),

          BIC(model_logit_antioxidants), BIC(model_logit_lipid_pero),

          BIC(model_logit_all))

)
```

```
print("Model Comparison (AIC/BIC):")
print(model_comparison)

# Stepwise variable selection on the full model
step_model <- step(model_logit_all, direction = "both", trace = 0)
summary(step_model)

# Select best model based on lowest AIC
best_model_name <- model_comparison$Model[which.min(model_comparison$AIC)]

# Predict using step model
data_complete$pred_prob <- predict(step_model, type = "response")
data_complete$pred_class <- ifelse(data_complete$pred_prob > 0.5, 1, 0)

# Confusion matrix
print("Confusion Matrix:")
confusion_mat <- table(Actual = data_complete$outcome, Predicted =
data_complete$pred_class)
print(confusion_mat)

# Accuracy
accuracy <- sum(diag(confusion_mat)) / sum(confusion_mat)
print(paste("Accuracy:", round(accuracy * 100, 2), "%"))

# ROC Curve and AUC
roc_curve <- roc(data_complete$outcome, data_complete$pred_prob)
plot(roc_curve, main = "ROC-Curve (Best Model)")
auc_value <- auc(roc_curve)
print(paste("AUC:", round(auc_value, 3)))

# Odds ratios for the best model (step_model)
```

```
print("Odds Ratios for Best Model:")

OR_table <- exp(cbind(OR = coef(step_model), confint(step_model)))

print(round(OR_table, 3))
```

Conclusion

Model Comparison:

The All_Variables model has the lowest AIC (733.64), indicating the best balance between fit and complexity among the full models. Stepwise selection further reduced AIC to 710, selecting 6 key predictors: lipids2, hormone1, hormone8, antioxidant4, antioxidant5, and lipid_pero3.

Significant Predictors:

hormone8, antioxidant4, and lipids2 were statistically significant ($p < 0.05$). hormone1, antioxidant5, and lipid_pero3 showed borderline significance.

Model Performance:

- Accuracy of 84.98% indicates the model correctly classifies most cases.
- Sensitivity is low (2/132 = 1.5%), meaning the model fails to identify most positive outcomes (outcome = 1).
- Specificity is high (739/740 = 99.9%), meaning it correctly identifies negatives.
- AUC of 0.67 indicates poor to fair discriminatory ability.

Odds Ratios Interpretation:

- lipids2 (OR = 1.385): For each unit increase in lipids2, odds of outcome = 1 increase by 38.5%.
- hormone1 (OR = 0.835): For each unit increase in hormone1, odds of outcome = 1 decrease by 16.5%.
- hormone8 (OR = 0.996): For each unit increase in hormone8, odds of outcome = 1 decrease by 0.4%.
- antioxidant4 (OR = 1.065): For each unit increase in antioxidant4, odds of outcome = 1 increase by 6.5%.
- antioxidant5 (OR = 1.806): For each unit increase in antioxidant5, odds of outcome = 1 increase by 80.6%.
- lipid_pero3 (OR = 1.532): For each unit increase in lipid_pero3, odds of outcome = 1 increase by 53.2%.

Overall Summary:

The logistic regression model has good accuracy but poor sensitivity, indicating it is better at ruling out the outcome than identifying it. hormone8 and lipids2 are the strongest predictors of the outcome.